

```
In [54]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

In [55]: df = pd.read_csv('data.csv')

In [56]: df.head()

Out[56]:
```

	#	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7	Unnamed: 8	Unnamed: 9
0	Session primary channel group (Default channel...	Date + hour (YYYYMMDDHH)	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session	Engagement rate	Event count
1	Direct	2024041623	237	300	144	47.52666666666700	0.6075949367088610	4.673333333333300	0.48	1402
2	Organic Social	2024041719	208	267	132	32.09737827715360	0.6346153846153850	4.296880149812730	0.4943820224719100	1147
3	Direct	2024041723	188	233	115	39.93991416309010	0.6117021276595740	4.587982832618030	0.49356223175965700	1069
4	Organic Social	2024041718	187	256	125	32.16015625	0.6684491978609630	4.078125	0.48828125	1044

```
In [57]: df.columns = df.iloc[0]
df = df.drop(index = 0).reset_index(drop = True)
df.columns = ["channel_group", "DateHour", "Users", "Sessions", "Engaged Session", "Average engagement time per session", "Engaged sessions per user", "Events per session", "Engagement rate", "Event count"]

In [58]: df.head()

Out[58]:
```

channel_group	DateHour	Users	Sessions	Engaged Session	Average engagement time per session	Engaged sessions per user	Events per session	Engagement rate	Event count	
0	Direct	2024041623	237	300	144	47.52666666666700	0.6075949367088610	4.673333333333300	0.48	1402
1	Organic Social	2024041719	208	267	132	32.09737827715360	0.6346153846153850	4.296880149812730	0.4943820224719100	1147
2	Direct	2024041723	188	233	115	39.93991416309010	0.6117021276595740	4.587982832618030	0.49356223175965700	1069
3	Organic Social	2024041718	187	256	125	32.16015625	0.6684491978609630	4.078125	0.48828125	1044
4	Organic Social	2024041720	175	221	112	46.918552036199100	0.64	4.529411764705880	0.5067873303167420	1001

```
In [59]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3182 entries, 0 to 3181
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  --
0   channel_group          3182 non-null   object
1   DateHour               3182 non-null   object
2   Users                  3182 non-null   object
3   Sessions               3182 non-null   object
4   Engaged Session        3182 non-null   object
5   Average engagement time per session  3182 non-null   float64
6   Engaged sessions per user  3182 non-null   float64
7   Events per session      3182 non-null   float64
8   Engagement rate         3182 non-null   float64
9   Event count            3182 non-null   object
dtypes: object(10)
memory usage: 249.7+ KB
```

cleaning data and data validation for the above data set

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In [61]: df.head()

Out[61]:
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channel_group	DateHour	Users	Sessions	Engaged Session	Average engagement time per session	Engaged sessions per user	Events per session	Engagement rate	Event count	
0	Direct	2024041623	237	300	144	47.52666666666700	0.6075949367088610	4.673333333333300	0.48	1402
1	Organic Social	2024041719	208	267	132	32.09737827715360	0.6346153846153850	4.296880149812730	0.4943820224719100	1147
2	Direct	2024041723	188	233	115	39.93991416309010	0.6117021276595740	4.587982832618030	0.49356223175965700	1069
3	Organic Social	2024041718	187	256	125	32.16015625	0.6684491978609630	4.078125	0.48828125	1044
4	Organic Social	2024041720	175	221	112	46.918552036199100	0.64	4.529411764705880	0.5067873303167420	1001

```
In [62]: df["DateHour"] = pd.to_datetime(df["DateHour"], format="%Y%m%d%H", errors="coerce")
numeric_cols = df.columns.drop(["channel_group", "DateHour"])
df[numeric_cols] = df[numeric_cols].apply(pd.to_numeric, errors="coerce")
df["Hour"] = df["DateHour"].dt.hour

In [64]: df.head()

Out[64]:
```

channel_group	DateHour	Users	Sessions	Engaged Session	Average engagement time per session	Engaged sessions per user	Events per session	Engagement rate	Event count	Hour	
0	Direct	2024-04-16 23:00:00	237	300	144	47.526667	0.607595	4.673333	0.480000	1402	23
1	Organic Social	2024-04-17 19:00:00	208	267	132	32.097378	0.634615	4.296880	0.494382	1147	19
2	Direct	2024-04-17 23:00:00	188	233	115	39.939914	0.611702	4.587983	0.493562	1069	23
3	Organic Social	2024-04-17 18:00:00	187	256	125	32.160156	0.668449	4.078125	0.488281	1044	18
4	Organic Social	2024-04-17 20:00:00	175	221	112	46.918552	0.640000	4.529412	0.506787	1001	20

```
In [65]: df.info()

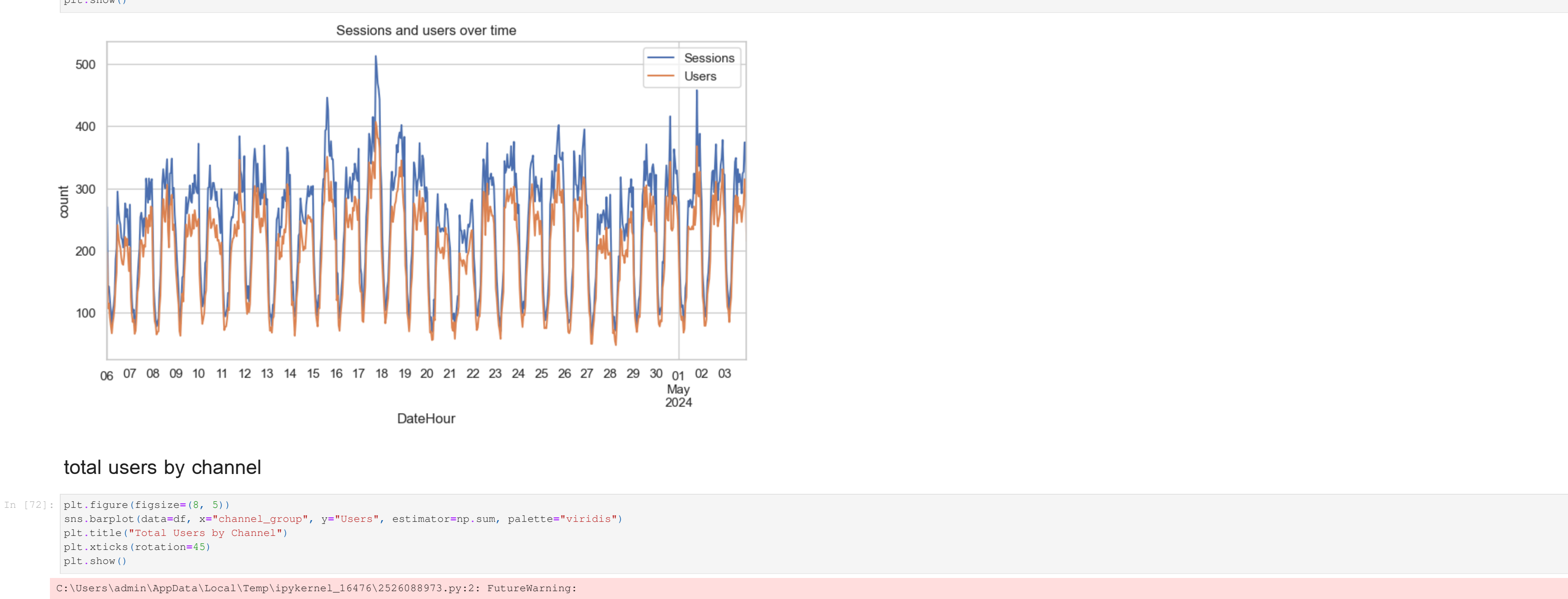
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3182 entries, 0 to 3181
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  --
0   channel_group          3182 non-null   object
1   DateHour               3182 non-null   datetime64[ns]
2   Users                  3182 non-null   int64
3   Sessions               3182 non-null   int64
4   Engaged Session        3182 non-null   int64
5   Average engagement time per session  3182 non-null   float64
6   Engaged sessions per user  3182 non-null   float64
7   Events per session      3182 non-null   float64
8   Engagement rate         3182 non-null   float64
9   Event count            3182 non-null   int64
10  Hour                   3182 non-null   int32
dtypes: datetime64[ns](1), float64(4), int32(1), int64(4), object(1)
memory usage: 261.2+ KB

In [66]: df.describe()

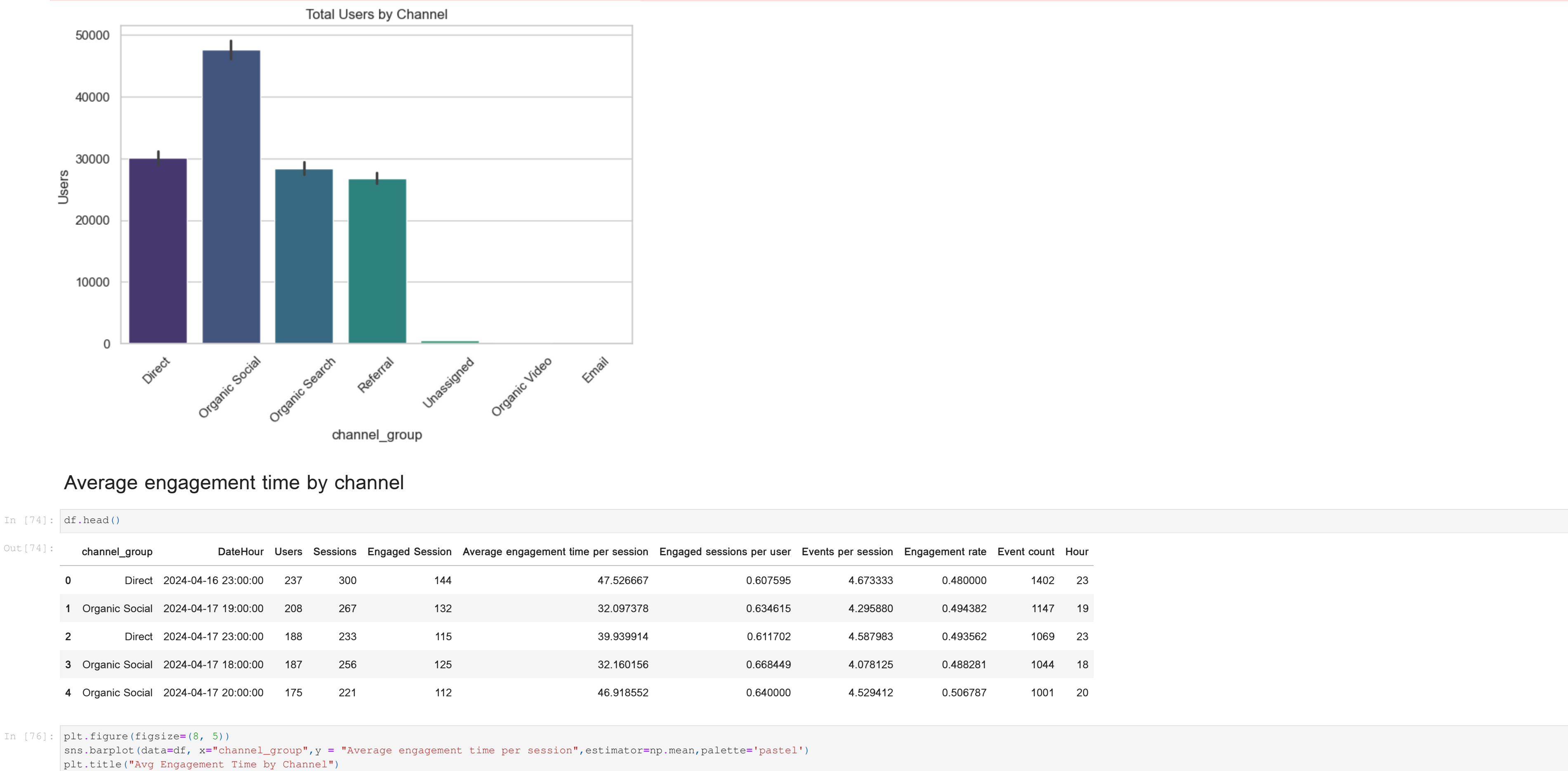
Out[66]:
```

	DateHour	Users	Sessions	Engaged Session	Average engagement time per session	Engaged sessions per user	Events per session	Engagement rate	Event count	Hour
count	3182	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000
mean	2024-04-20 01:17:07.28441216	41.935889	51.192846	28.325681	66.644581	0.606450	4.675969	0.503396	242.272470	11.807040
min	2024-04-06 00:00:00	0.000000	1.000000	0.000000	0.000000	0.000000	1.000000	0.000000	0.000000	0.000000
25%	2024-04-13 02:15:00	20.000000	24.000000	13.000000	32.102034	0.561404	3.750000	0.442902	103.000000	6.000000
50%	2024-04-20 02:00:00	42.000000	51.000000	27.000000	49.020202	0.666667	4.410256	0.545455	226.000000	12.000000
75%	2024-04-26 22:00:00	60.000000	71.000000	41.000000	71.487069	0.750000	5.217690	0.633333	339.000000	18.000000
max	2024-05-03 23:00:00	237.000000	300.000000	144.000000	4525.000000	2.000000	56.000000	1.000000	1402.000000	23.000000
std	NaN	29.582298	36.919962	20.650969	127.200659	0.264023	2.795228	0.228206	184.440313	6.886696

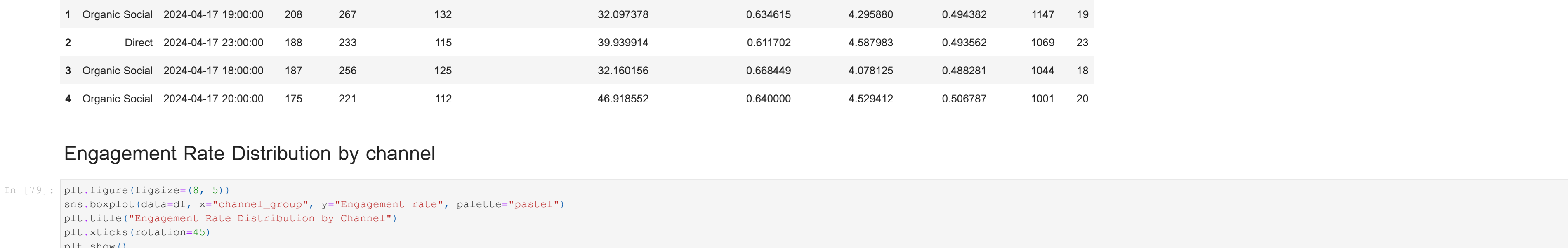
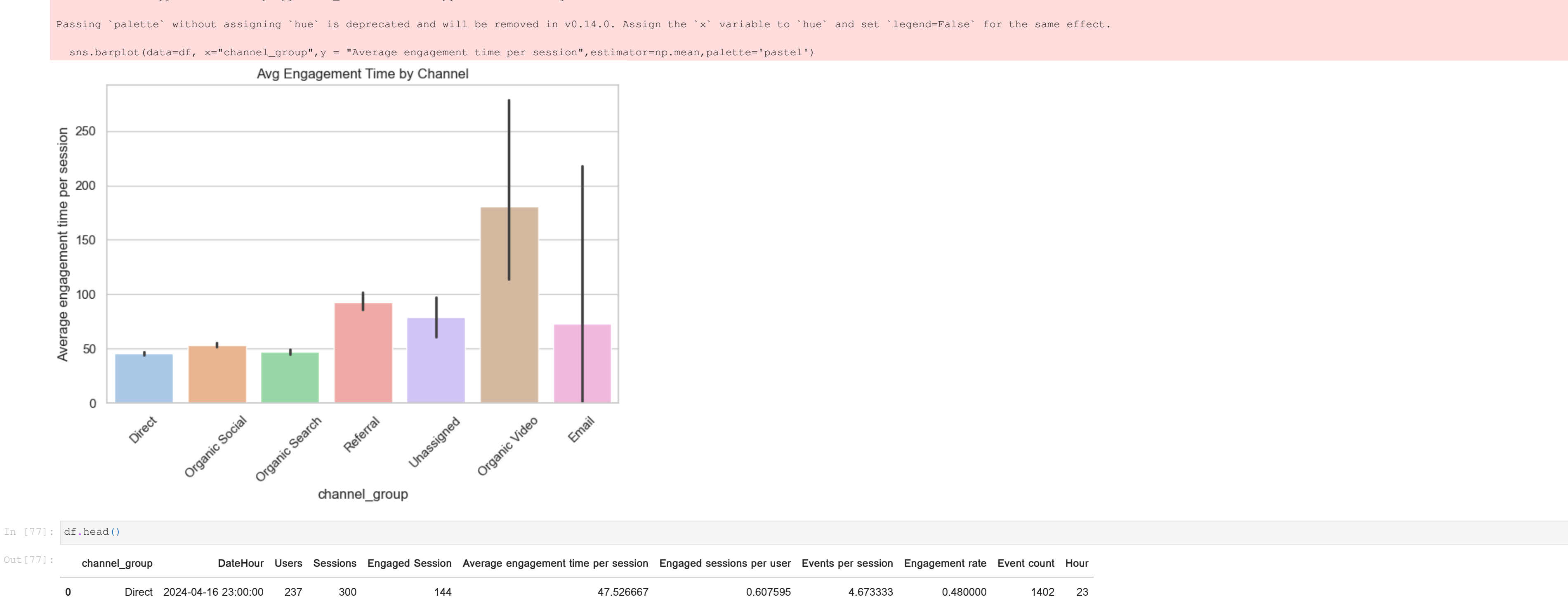
sessions and user over time



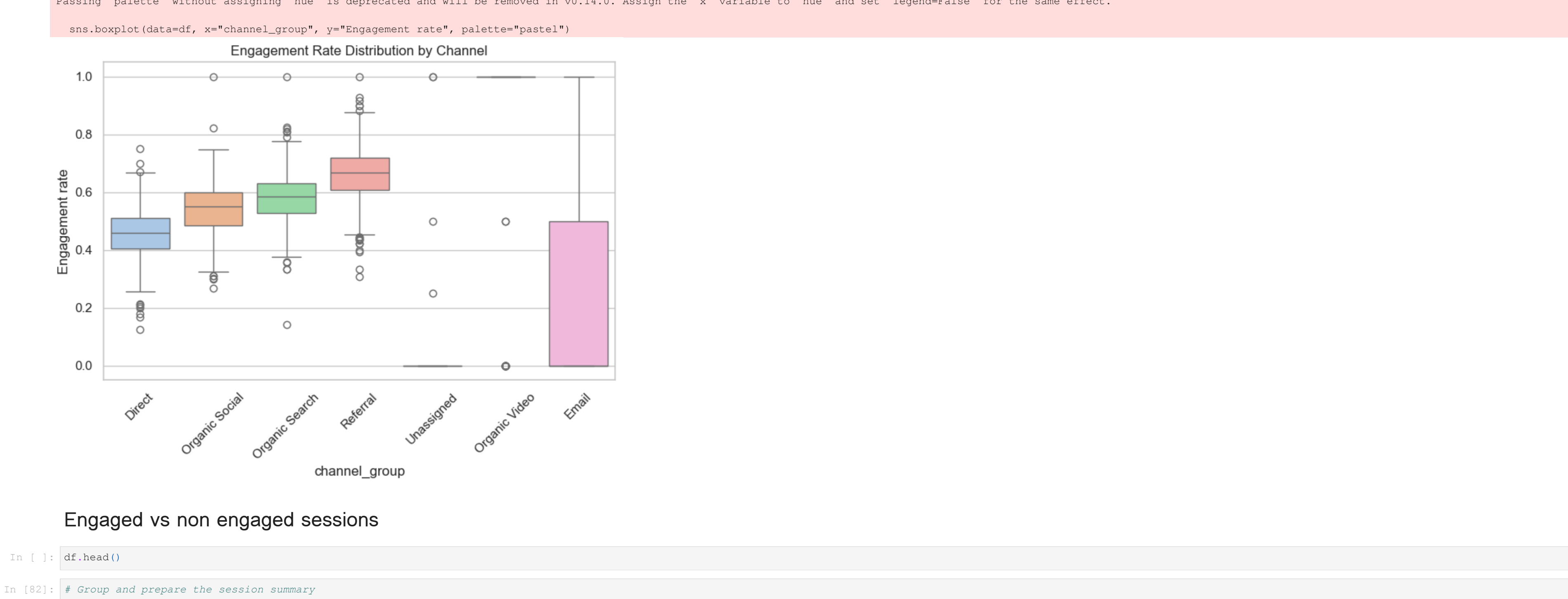
total users by channel



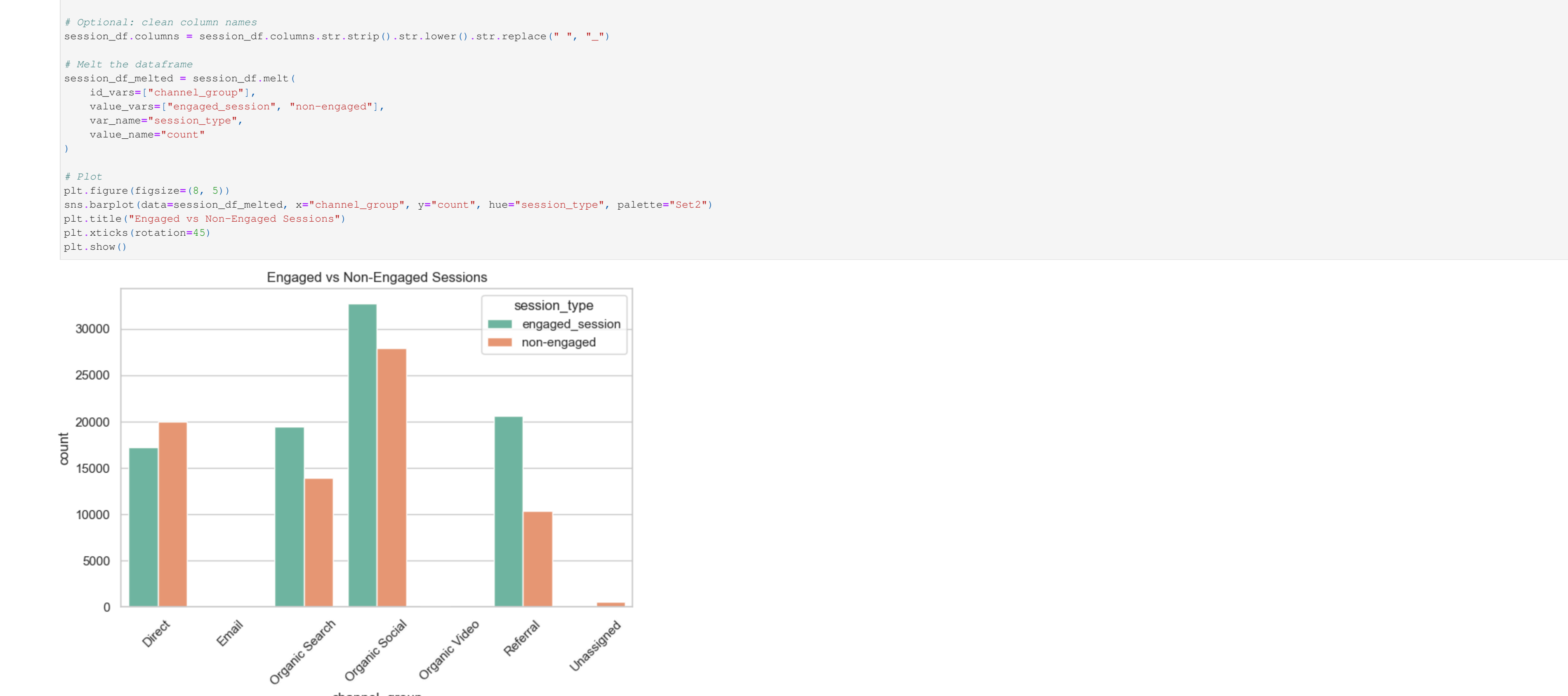
Average engagement time by channel



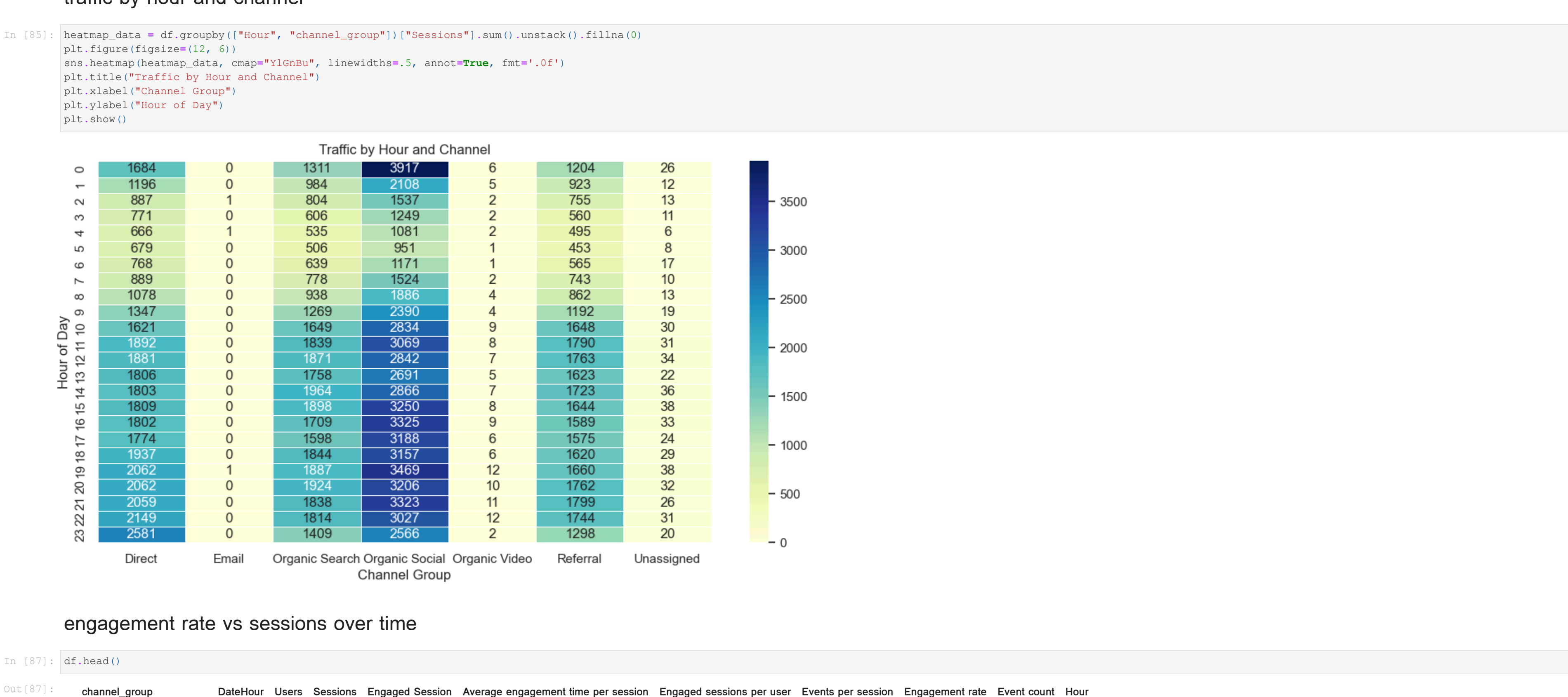
Engagement Rate Distribution by channel



Engaged vs non engaged sessions



traffic by hour and channel



engagement rate vs sessions over time

